

AI Video Analytics & ByteTrack Tracking Report

Video File: video_1785754221482_92480444.mp4 | Generated: 2026-08-03 16:21:15

1. Executive Summary

Automated AI vision tracking using YOLOv8 + ByteTrack was completed for **video_1785754221482_92480444.mp4**. The video was processed across **311 frames** (Duration: 12s, Resolution: 768x432). A total of **51 unique people** were assigned persistent tracking IDs. Maximum simultaneous occupancy reached **5** with an average occupancy of **2.4**. Crowd level was evaluated at **Very Low (10.0%)** based on a store capacity benchmark. Total footfall registered **18** (Entries: 6, Exits: 12) with an average detection confidence of **77.4%**.

2. Key Analytics Summary Table

Metric Parameter	Recorded Value	Benchmark / Context
Unique Tracked People	51	ByteTrack persistent unique tracking IDs
Current Occupancy	2	Visible tracked individuals in final frame
Maximum Occupancy (Peak)	5	Highest simultaneous tracked count
Minimum Occupancy	0	Lowest simultaneous tracked count
Average Occupancy	2.4	Mean tracked individuals per frame
Crowd Occupancy Rate	10.0%	Density Level: Very Low
Total Footfall Count	18	Entries: 6 Exits: 12
Average AI Confidence	77.4%	Range: 50.2% - 94.1%
Total Frames Processed	311	FPS: 25.0 Resolution: 768x432
Processing Execution Time	53.39 seconds	Real YOLOv8 + ByteTrack runtime

3. Object Class Statistics

Category	Count	Percentage
Person (Tracked)	51	100.0%

4. AI Insights & Recommendations

"Automated YOLOv8 + ByteTrack video analysis completed for video_1785754221482_92480444.mp4 (12s). Tracked 51 unique individuals across 311 frames with ByteTrack persistent ID mapping. Peak simultaneous occupancy reached 5 (Avg occupancy: 2.4). Crowd density evaluated at 10.0% (Very Low Status) relative to store capacity (50). Registered 6 entries and 12 exits with an average detection confidence of 77.4%."

- Unique shoppers tracked: 51. Ensure shelf layout maximizes engagement for peak traffic zones.
- Current crowd density level is Very Low (10.0%). Align staff allocation with peak occupancy windows.
- Recorded 6 entrance line crossings and 12 exit crossings.
- ByteTrack persistent tracking maintained across 311 frames with 77.4% average confidence.